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Morphometrics and genetics highlight the complex history of Eastern Mediterranean spiny mice

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Spiny mice of the *Acomys cahirinus* group display a complex geographical structure in the Eastern Mediterranean area, as shown by previous genetic and chromosomal studies. To better elucidate the evolutionary relationships between insular populations from Crete and Cyprus and continental populations from North Africa and Cilicia in Turkey, genetic and morphometric variations were investigated, based on mitochondrial D-loop sequences, and the size and shape of the first upper molar. The Cypriot and the Cilician populations show idiosyncratic divergence in molar size and shape, while Cretan populations present a geographical structure with at least three differentiated subpopulations, as shown by congruent distributions of haplogroups, Robertsonian fusions and morphometric variation. A complex history of multiple introductions is probably responsible for this structure, and insular isolation coupled with habitat shift should have further promoted a pronounced and rapid morphological evolution in molar size and shape on Crete and Cyprus.

ADDITIONAL KEYWORDS: *Acomys minous* – *Acomys cilicicus* – *Acomys nesiotes* – Crete – Cyprus – D-loop – geometric morphometrics – insular evolution – molar shape – phylogeography.

INTRODUCTION

Spiny mice of the genus *Acomys* occupy a large distribution area over Africa to Western Asia, including a small area along the Mediterranean coast of Turkey, a region historically known as Cilicia. They further occur on the islands of Crete and Cyprus, within the Eastern Mediterranean area, which along with Cilicia constitute the northernmost distribution limit of the genus. The lack of clear diagnostic characters has led to taxonomic debate regarding a group of closely related species known as the *cahirinus–dimidiatus* group (Denys *et al.*, 1994; Volobouev *et al.*, 2007; Frynta *et al.*, 2010; Aghová *et al.*, 2019). A growing body of genetic and chromosomal evidence has shown that one clade, attributed to *A. dimidiatus*, now occupies the Levant, including Israel and Sinai, up to Arabia and Iran. The other clade (*A. cahirinus*) would thus

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only occur between the Eastern Sahara and Egypt along the Nile Valley (Volobouev *et al.*, 2007; Frynta *et al.*, 2010). Spiny mice from Crete, Cyprus and Cilicia are affiliated to the second group (Barome *et al.*, 2001; Frynta *et al.*, 2010; Giagia-Athanasopoulou *et al.*, 2011). As they are considered endemics, some displaying distinct morphological features, such as a larger body size (Kryštufek & Vohralík, 2009), each of these populations has been given a specific or subspecific status: *A. nesiotes* for the Cypriot, *A. cilicicus* for the Cilician and *A. minous* for the Cretan populations. Crete, however, is characterized by genetic heterogeneity, with the coexistence of two distinct mitochondrial clades, one also found in Cyprus and the low Nile valley (Cairo, Egypt), and the other one in Cilicia (Barome *et al.*, 2001) and the high Nile Valley, Libya and Chad (Frynta *et al.*, 2010). Cretan mice further vary in their chromosomal number, which ranges from $2n = 38$ to $2n = 42$, due to Robertsonian (Rb) fusions. No individual association has been found between chromosomal number and mitochondrial lineage in these mice (Giagia-Athanasopoulou *et al.*, 2011).

To better understand the evolutionary history of the *cahirinus* '*sensu lato*' group (including *A. minous*, *A. nesiotes* and *A. cilicicus*), a geometric morphometric analysis of the first upper molar (Fig. 1) was performed. In *Acomys* as in murine rodents, the first molar erupts before weaning and is subsequently affected only by wear. Molar shape differences are thus indicative of underlying genetic changes. In contrast, osteological characters, such as skull or mandible, are prone to plastic remodelling along an animal's life and consequently they also vary on a short timescale, depending on local ecological conditions (Caumul & Polly, 2005; Ledevin *et al.*, 2012). Furthermore, molar teeth are the most frequent fossil remains of small mammals; their study allows a comparison of modern and ancient samples, providing a temporal perspective. Hence, the morphometric analysis of molar shape may deliver valuable insight into the genetic differentiation among populations, complementary to the analysis of mitochondrial DNA, for which existing data were supplemented with new mitochondrial D-loop sequences from mice from Crete, Cyprus and Cilicia.

Based on these morphometric and genetic data, the following questions were addressed: (1) Is the genetic and chromosomal heterogeneity on Crete mirrored in high morphological disparity? (2) Can a synthesis of the genetic, morphometric and chromosomal data shed light on the processes that led to the diversity within *Acomys cahirinus s.l.* in the Eastern Mediterranean area? (3) Is insularity or isolation the prime driver of morphological divergence? If drift in isolated populations is the dominant factor, a pronounced morphological divergence is expected in Cilicia, as on the islands of Crete and Cyprus. If adaptation to island-specific ecological conditions, including changes in the level of competition and predation, is the prime driving factor of divergence (Lomolino, 1985, 2005), the Cilician population should display less divergence compared to those of Crete and Cyprus.

MATERIAL AND METHODS

Sampling for genetics

Thirty new D-loop sequences were acquired, including specimens from Cyprus, Crete and Cilicia (Table 1): (1) five Cypriot specimens ('*A. nesiotes*') collected in 2015 – despite sampling effort around the island, specimens were only caught on Cap Greco or adjacent to it; (2) 11 Cretan specimens ('*A. minous*'); and (3) 14 Cilician individuals ('*A. cilicicus*') trapped in spring 2013, around Narlıkuyu, in the district of Mersin (Turkey).

Sampling for morphometrics

The morphometric study (Table 1) included 92 specimens collected in Crete (61), Cyprus (six) and Cilicia (17). This sampling was complemented by eight specimens from Cairo (Egypt) from the Museum National d'Histoire Naturelle de Paris (vouchers: 2001-11; 1997-1308; 1996-432; 1996-446; 1996-431; 1996-430; 1994-1280; 1999-6), identified as *A. cahirinus*.

Four other specimens from the Museum National d'Histoire Naturelle de Paris were attributed with less certainty to *A. cahirinus*, but coming from Sudan and Chad, they allowed us to document the geographical variation at a larger scale (vouchers: 1906-118a; 1906-118b; 1906-118c; 1981-1059).

Body size and sex data were available for most specimens, allowing us to investigate the occurrence of sexual dimorphism and allometry. All specimens were identified as sub-adult or adult, based on the criterion of the eruption of the third molars. One specimen from Cairo (Egypt) was a juvenile, as its third molars were not erupted. It was not included in the analyses of body size, but because the molar does not grow further after its eruption, it was included in the analyses of first molar size and shape.

Finally, three fossil teeth from the floor of the Hellenistic Temple C at the ancient port of Kommos on the south coast of Crete, dated to between 375 BC and AD 160/170 (Shaw, 2000), were measured based on drawings of published plates (Payne, 1995) and included in the morphometric analyses.

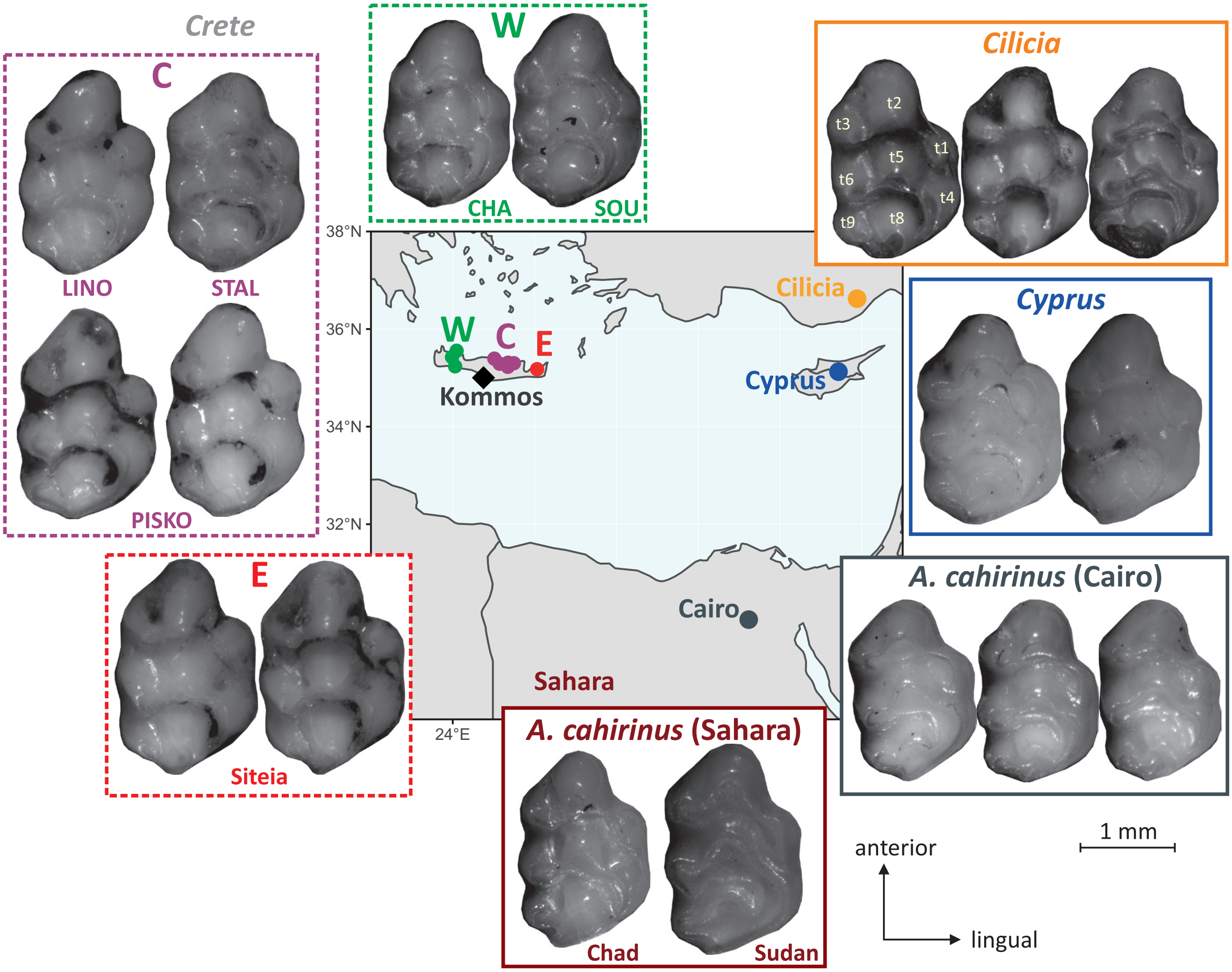

Figure 1. Sampling localities and examples of first upper molars (UM1) in the various populations of *Acomys* considered in this study. All belong to the *cahirinus s.l.* group. Spiny mice from the island of Crete are designated as *A. minous*, those from the island of Cyprus as *A. nesiotes* and those confined to a small area in Cilicia (Asia Minor, Turkey) as *A. cilicicus*. Nomenclature of the cusps is indicated on one molar tooth of *A. cilicicus*. The boxes indicate the colour code of the areas in figures for morphometrics. For abbreviations of some localities (LINO, STAL, PISKO) see Table 1. All teeth are to the same scale (scale bar bottom right).

Genetic analyses

Acomys tissue samples were extracted using DNEasy from Qiagen following the manufacturer instructions. A D-loop fragment of 514 bp was amplified using previously described primers (Nicolas *et al.*, 2009). PCR conditions were as follows: 10 ng DNA, 2 mM $MgCl_2$, 0.2 mM dNTP, 0.5 U Taq and 0.2 µM of each primer. PCR cycles were 15 min at 95 °C, followed by 35 cycles of 30 s at 95 °C, 90 s at 54 °C and 1 min at 72 °C, with a final elongation of 15 min at 72 °C. The sequences generated were visualized and analysed using the CLC Workbench (Qiagen) and aligned with Seaview v.4 (Gouy *et al.*, 2010). Genetic diversity indices were calculated using DNAsp (Librado & Rozas, 2009). Differentiation among populations was assessed using Fst values calculated with Arlequin (Excoffier & Lischer, 2010).

A phylogenetic tree was generated using MrBayes (Ronquist *et al.*, 2012) and PhyML 3.0 (Guindon *et al.*, 2010), including the 30 sequences generated in the present study and the 16 sequences available from GenBank, leading to a total of 46 sequences of *A. cahirinus s.l.* The haplotypes were determined with DNAsp (Librado & Rozas, 2009). Only the 20 haplotypes differing by mutation were retained for phylogenetic analysis. We added to these haplotypes seven *A. dimidiatus* (AJ012028, MH044889, MH044888, MH044871, MH044868, MH044840, FJ415545) as well as three *A. ignitus* (MH044872, MH044875, MH044876), two *A. wilsoni* (MH044862, MH044874)

Table 1. Sampling for morphometric and genetic analyses

Area	Region	Locality	Code	*N* UM1	*N* D-loop	New accession numbers
Crete	West (W)	Chania – Akrotirio	CHA	5	**1**	MT001851
		Lefka Ori	LEFKO	5	**1**	MT001845
		Chania – Souda	SOU	3	**1**	MT001848
		Hills facing Elafonisos island			1	
		Kournas Lake			1	
	Central (C)	Almyros Gorge, Linoperamata	LINO	6	**1**	MT001852
		Kokkini Hani	KOKH	8	**1**	MT001844
		Stalida Mochos	STAL	3	**1**	MT001846
		Piskopiano	PISKO	5	**2**	MT001847 MT043301
		Towards Kokkini Chani	PKOK	20	**1**	MT001853
	East (E)	Siteia	SIT	6	**2**	MT001849 MT001850
		Vai			2	
Cyprus		Cap Greco	CYP	6	**5**	MT001854-58
		Agirdag			1	
		Cinarli			1	
		Zafer Burnu, Cap Andreas			1	
Turkey	Cilicia (CIL)	Narlıkuyu	NAR	6	**6**	MT001830-35
		Ayaş	AYA	4	**3**	MT001836-38
		Karaahmetli, Hüseyinler, Kumrukuyu	KAR	7	**5**	MT001839-43
		Silifke			2	
		NA			1	
Libya		Mts Akakus			1	
Egypt	Cairo	Cairo	CAIRO	8	2	
	Assouan	Abu Simbel			2	
Sahara	Sahara	Sudan (Kharthoum)	SAH	3		
		Chad (Yogoum)	SAH	1		
		Chad (Tibesti plateau)			1	
Fossil Crete	Central Crete	Kommos	Fos-KOM	3		

Region, area within region, locality of trapping and its code are provided. *N* UM1, number of first upper molars included in the analysis. *N* D-loop, number of D-loop sequences in the present study.

and three *A. russatus* (MH044881, MH044885 and FJ415546) that were used as outgroups. Sites with more than 20% missing data were removed and the final alignment comprised 35 sequences and 493 sites. The substitution model, GTR+I+G, was chosen using jmodeltest (Darriba *et al.*, 2012). Robustness of the nodes was estimated with 1000 bootstrap replicates with PhyML and posterior probability with MrBayes. The generation number was set at 2 000 000 Markov chain Monte Carlo replications with one tree sampled every 500 generations. The burn-in was graphically determined with Tracer v.1.7 (Rambaut *et al.*, 2018). We also checked that the effective sample sizes (ESSs) were above 200 and that the average standard deviation of split frequencies remained <0.01 after the burn-in threshold. We discarded 20% of the trees and visualized the resulting tree with Figtree v.1.4 (Rambaut, 2012). A median-joining haplotype network was constructed in PopART (Leigh & Bryant, 2015) with the 46 *A. cahirinus* sequences only. All the sequences generated in the present study are available on GenBank (accession numbers MT001830–MT001858, MT043301) (Table 1).

To estimate the divergence time of the lineages of *A. cahirinus* in the Eastern Mediterranean area, we used BEAST v.2.5.2 (Bouckaert *et al.*, 2019), including its functions BEAUTI, LogCombiner and TreeAnnotator. Mitochondrial (*Cyt b* + D-loop) and nuclear genes (*IRBP*, *RAG1*) were retrieved from GenBank for 32 *Acomys* representing the main lineages within this genus (Aghová *et al.*, 2019) and four outgroups (*Deomys ferrugineus*, *Lophuromys flavopunctatus*, *Lophuromys sikapusi* and *Uranomys ruddi*) (Supporting Information Table S1). The two Eastern Mediterranean lineages were each represented by one specimen for which sequences of the four genes were available (Table S1). The dataset comprised 36 sequences and 3509 bp. The best partitioning scheme and substitution models were determined with PartitionFinder 2 (Lanfear *et al.*, 2017) using a greedy heuristic algorithm with 'linked branch lengths' option and the Bayesian Information Criterion (BIC) (Table S2). The partitions were imported into BEAUTI, where they were assigned separate and unlinked substitution and clock models. Bayesian analyses were run with uncorrelated lognormal relaxed clocks, birth–death tree prior and three fossil constraints defined by using lognormal statistical distributions. We used the fossil constraints proposed by Aghova *et al*. (2019) within the genus *Acomys*, and hence the same specifications of lognormal priors: offset of 8.5 Mya for the most recent common ancestor (MRCA) of the genus *Acomys*, 6.08 Mya for the MRCA of the clade encompassing *cahirinus* + *wilsoni* + *russatus* and 3 Mya for the MRCA of the *spinosissimus* group (Aghová *et al.*, 2019); mean = 1 in the three cases (Aghová *et al.*, 2019). Two independent runs were carried out for 50 million generations with sampling every 1000 generations in BEAST. The first 10% were discarded as burn-in and the resulting parameter and tree files were examined for convergence and ESSs in Tracer 1.7 (Rambaut *et al.*, 2018). The two runs were combined in LogCombiner and the species tree was summarized with TreeAnnotator and visualized with Figtree v.1.4 (Rambaut, 2012).

Morphometrics analyses

First upper molars (UM1) were photographed using a Leica MZ 9.5 binocular microscope, being manually orientated so that the occlusal surface matched at best the horizontal plane. The shape of UM1 was described using 64 points sampled at equal curvilinear distance along the two-dimensional outline of the occlusal surface using the Optimas software. An outline-based method was chosen, because reliable landmarks are difficult to position on murine-like molars. The top of the cusps is abraded by wear and cannot be used to assess the position of the cusps, and landmarks bracketing the cusps on the outline are difficult to position, given the smooth undulations delineating the cusps along the outline. The starting point was tentatively positioned at the anterior-most part of the tooth.

The points along the outline were analysed as sliding semi-landmarks (Cucchi *et al.*, 2013). Using this approach, the outline points are adjusted using a generalized Procrustes superimposition (GPA) standardizing size, position and orientation, while retaining the geometric relationships between specimens (Rohlf & Slice, 1990). During the superimposition, semi-landmarks were allowed sliding along their tangent vectors until their positions minimized the shape difference between specimens, the criterion being bending energy (Bookstein, 1997). Because the first point was only defined on the basis of a maximum of curvature at the anterior-most part of UM1, some slight offset might occur between specimens. The first point was therefore considered as a semi-landmark allowed to slide between the last and second points.

The centroid size (CS) of the 64 points (i.e. square root of the sum of the squared distance from each point to the centroid of the configuration) was considered as an estimate of overall tooth size. Differences between groups were tested using an analysis of variance (ANOVA) and relationships between variables were assessed using Pearson's product-moment correlation. The pattern of tooth shape differentiation was explored using multivariate analyses of the aligned coordinates. A principal component analysis (PCA on the variance–covariance matrix of the aligned coordinates) allowed a first exploration of the dataset. It was complemented by a between-group PCA (bgPCA). While the PCA is an eigenanalysis of the total variance–covariance of the dataset, the bgPCA analyses the variance–covariance between group means weighted by the sample size of each group.

Size-related variations in shape and differences between groups were investigated using Procrustes ANOVA. With this approach, the Procrustes distances among specimens are used to quantify the components of shape variation, which are statistically evaluated via permutation (here, 9999 permutations) (Adams & Otarola-Castillo, 2013). The allometric relationship was visualized as the common allometric component (CAC) (Adams *et al.*, 2013). To assess the impact of allometry on the pattern of shape differentiation, a bgPCA was also performed on the residuals of a regression of the aligned coordinates vs. centroid size.

The GPA and the Procrustes ANOVA were performed using the R package *geomorph* (Adams & Otarola-Castillo, 2013). The PCA and the bgPCA were performed using the R package *ade4* (Dray & Dufour, 2007). For ANOVA and Procrustes ANOVA, the grouping factor was considered to be the region (see Table 1), in order to increase sample size and ameliorate the performance of the tests. For the bgPCA, the grouping factor was the trapping locality. It corresponds to field data, and the clustering of neighbouring localities in the morphospace is indicative of geographical structure, despite low sample size for small groups.

RESULTS

Genetic analyses

As the phylogenetic trees reconstructed with MrBayes and PhyML were congruent, only the Bayesian tree is presented (Fig. 2). Two main lineages were found with good support [bootstrap percentage (BP) > 0.75, posterior probability (PP) > 0.45] within *A. cahirinus s.l.* Lineage A (according the terminology of Barome *et al.*, 2001) consisted of haplotypes from Western and Central Crete, Cyprus and Cairo (Northern Egypt). Lineage B involved samples from Eastern and Central Crete, Cilicia and Southern Egypt. The sequences of Chad and Libya appear to be basal with regard to

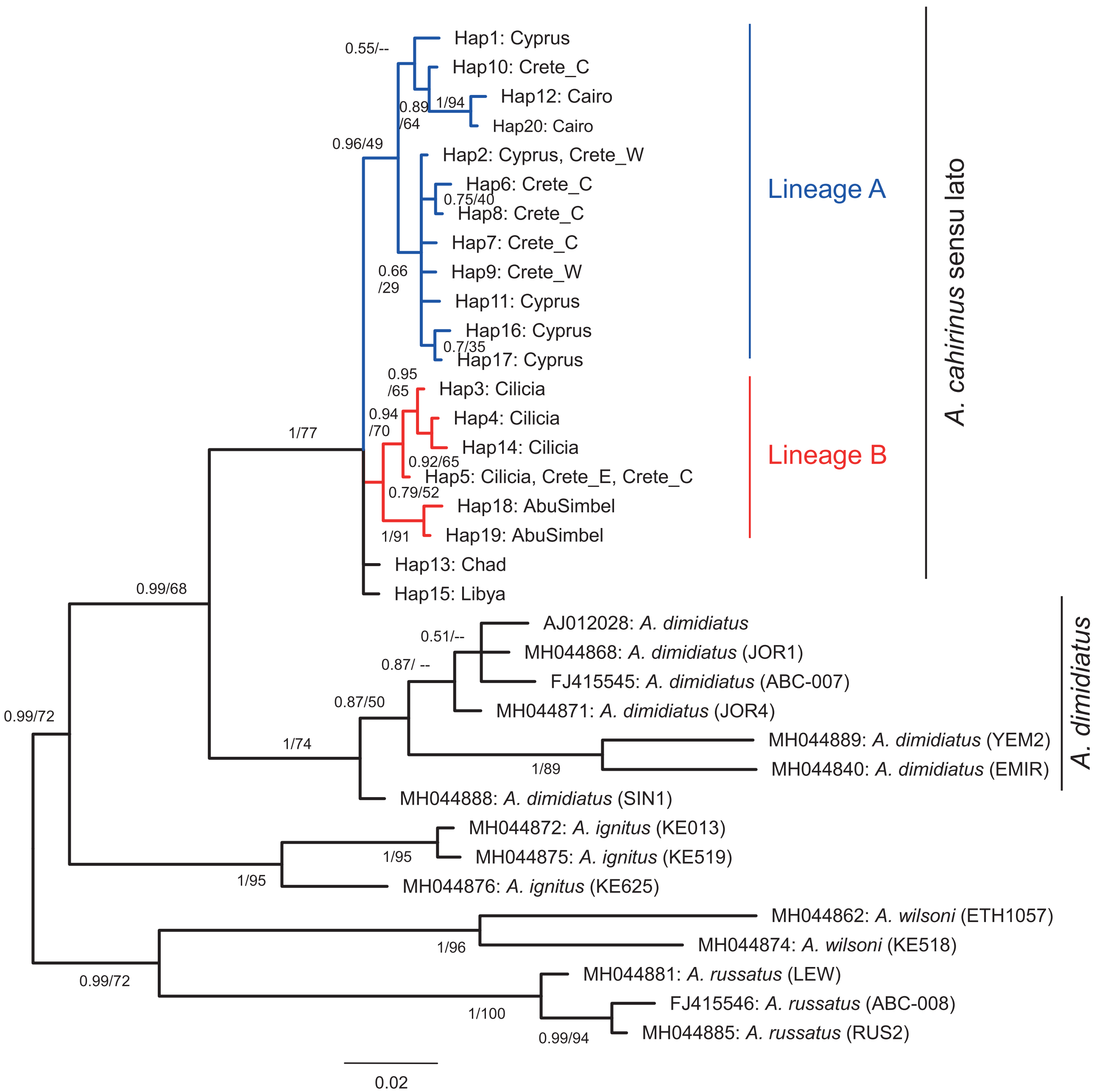

Figure 2. Bayesian phylogeny of D-loop haplotypes of *Acomys cahirinus s.l.* Posterior probability/bootstrap support values are indicated for each node. A dash indicates that the node is not supported in the phylogeny reconstructed with PhyML. Scale bar correspond to 0.2 substitutions per site.

the two lineages, and they were attributed neither to lineage A nor to lineage B.

Both D-loop and *Cyt b* showed two lineages within *A. cahirinus s.l.* and the two mitochondrial genes provided congruent outcomes at the individual level (Barome *et al.*, 2001; Frynta *et al.*, 2010; this study). One exception regarded specimen R155 from Piskopiano (Crete) published with a *Cyt b* sequence belonging to lineage B (Giagia-Athanasopoulou *et al.*, 2011) and for which we found a D-loop belonging to lineage A. We re-sequenced both genes and confirmed the D-loop attribution to lineage A. Another discrepancy regarded the North African specimens, which were attributed to lineage B for *Cyt b* (Frynta *et al.*, 2010). Based on the present D-loop results, they appear to have a basal position in the phylogenetic tree with regard to lineages A and B.

Eastern and Central Crete shared haplotypes with Cilicia. Western and Central Crete shared other

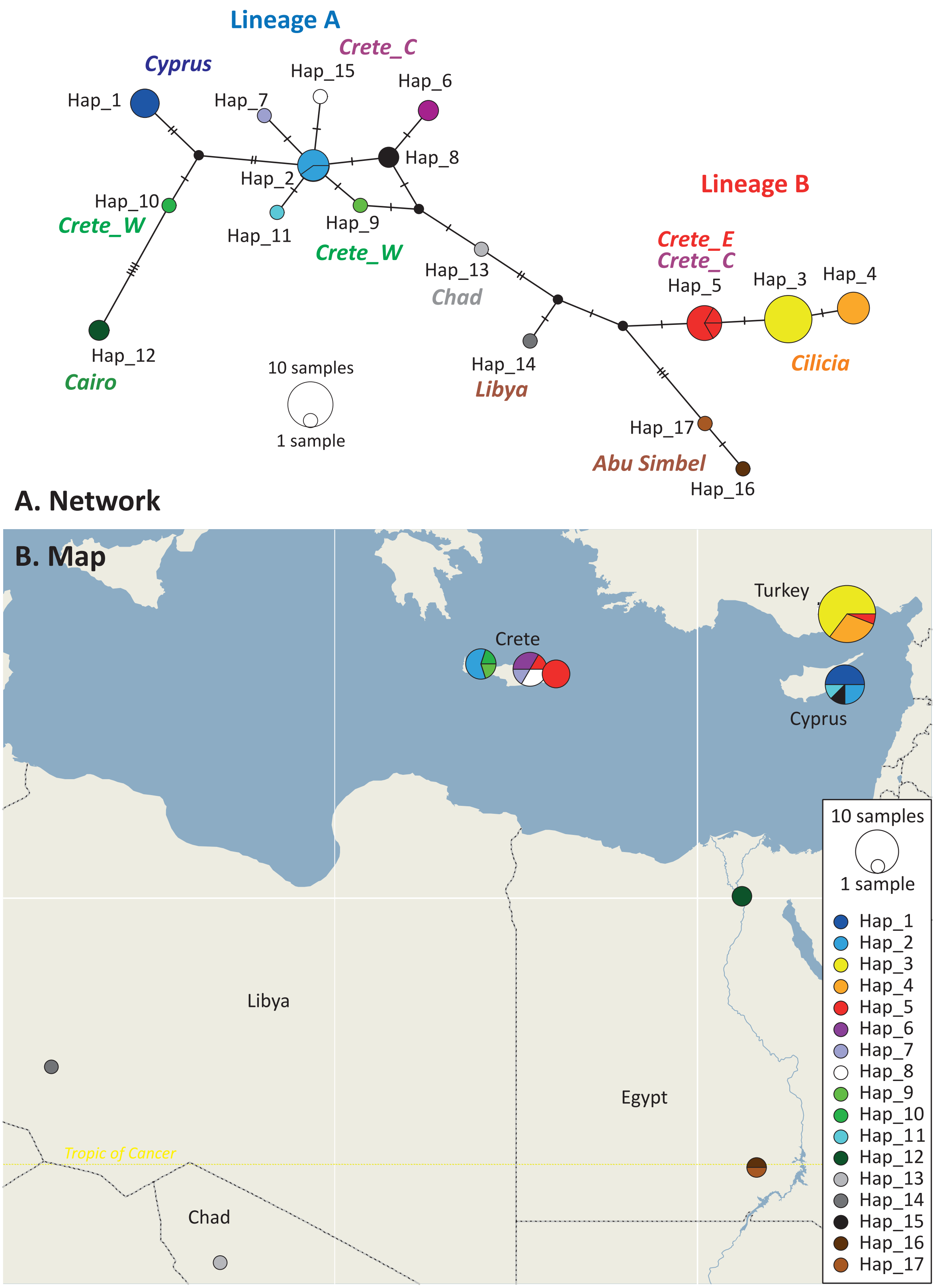

Figure 3. Network and geographical distribution of the D-loop haplotypes. A, median-joining haplotype network. B, geographical repartition of the haplotypes. Each haplotype is identified by the same colour code on both figures.

haplotypes with Cyprus; none were in common between Cyprus and Cilicia (Fig. 2). This result is confirmed by the F_{st} values being lower between Cyprus and Crete (F_{st} = 0.28, P = 0.01) and between Cilicia and Crete (F_{st} = 0.60, P < 0.001) than between Cyprus and Cilicia (F_{st} = 0.83, P < 0.001). Haplotype diversity (*Hd*) was found to be higher in Crete (*Hd* = 0.864) than in Cyprus (*Hd* = 0.667) or Cilicia (*Hd* = 0.448).

In the network (Fig. 3), a finer geographical structure could be recognized, especially in Crete. The few African samples appeared as central in the network, with the exception of those from Cairo, which appeared to be divergent but nested within lineage A. Haplotypes from Crete appeared intermediate between the basal African haplotypes and either those of Cyprus (for lineage A) or those of Cilicia (for lineage B). A geographical structure emerged in Crete, with different haplotypes found in the West, Centre and East of the island. Within lineage A, a central haplotype (in blue on Fig. 3) was found in both Western Crete and Cyprus; other haplotypes were characteristic of either Western or Central Crete. Eastern Crete was characterized by a haplotype affiliated to lineage B, also present in the easternmost locality of Central Crete (Stalida Mochos) and in Cilicia. Cilicia further hosted two specific haplotypes, while Cyprus displayed three specific haplotypes.

Based on a combined analysis of mitochondrial (*Cyt b* + D-loop) and nuclear genes (*IRBP*, *RAG1*), the divergence between the two lineages was estimated at 210 000 years (range 110 000–320 000 years) (Supporting Information, Fig. S1).

Morphometric analysis of differentiation in first upper molar

Sexual dimorphism

Data on sex and body size (head + body length) were available for most specimens. Body size, as well as UM1 centroid size, were well differentiated among regions, but not between sexes (Table 2). Tooth shape was also very different among regions, while sexes were only weakly differentiated. As a consequence, males and females were pooled in subsequent analyses, to focus on geographical patterns of differentiation.

Geographical variations in size, and relationship between body and tooth size

Body size varied greatly among but also within populations (Fig. 4A). This variation is partly due to the age structure of the populations, because very young animals can be trapped, as shown by the specimen without erupted third molars, which displays a very small body size. Specimens from Western Crete tended to display the largest body size, whereas those from Cilicia were among the smallest. Animals from Cyprus and Cairo tended to have a large body size.

This trend was not reflected in the pattern of molar size variation (Fig. 4B). Animals from Western Crete, while being among the largest, displayed small molar teeth. Similarly, the large animals from Cairo displayed extremely small teeth, while the spiny mice from Cyprus, of similar body size, displayed among the largest teeth of the dataset. As a consequence, body and molar size were not related (Fig. 4C). In a model including region and body size as explanatory variables, UM1 centroid size varied greatly among regions (P < 2e-16) but varied only weakly with body size (P = 0.0158).

Overall, tooth size displayed a reduced variation within localities and within regions, in contrast to body size, but the differences in molar size were important even among regions of Crete. Spiny mice from Saharan regions (Sudan and Chad) displayed important variations in molar size, even within Sudan, despite the fact that these mice were all derived from Khartoum. The fossil teeth from Kommos were large, similar to the modern Central and Eastern Cretan populations.

Tooth shape differences among populations

The different regions were highly different in shape (Procrustes ANOVA: P < 0.0001). In the morphospace defined by the first two axes of the PCA on the aligned coordinates (Fig. 5A), populations from Western and Eastern Crete were opposed along PC1. This axis describes a change from teeth with receding anterior cusps leading to an elongated anterior part, to teeth with prominent, anteriorly shifted lingual cusp t1 and labial cusp t3 (Fig. 5B). Negative scores along PC2

Table 2. Sexual dimorphism in body size and molar size and shape, taking into account the regions

Method	Variable(s)	Region	Sex	Interaction
ANOVA	HBL	**<0.0001**	0.8050	0.9425
	UM1 CS	**<0.0001**	0.1679	0.7630
ProcANOVA	UM1 Shape	**<0.0001**	0.0133	0.0028

Probabilities from ANOVA are given for size-univariate parameters (HBL, head + body length; UM1 CS: first upper molar centroid size) and from Procrustes ANOVA for shape data (aligned coordinates).

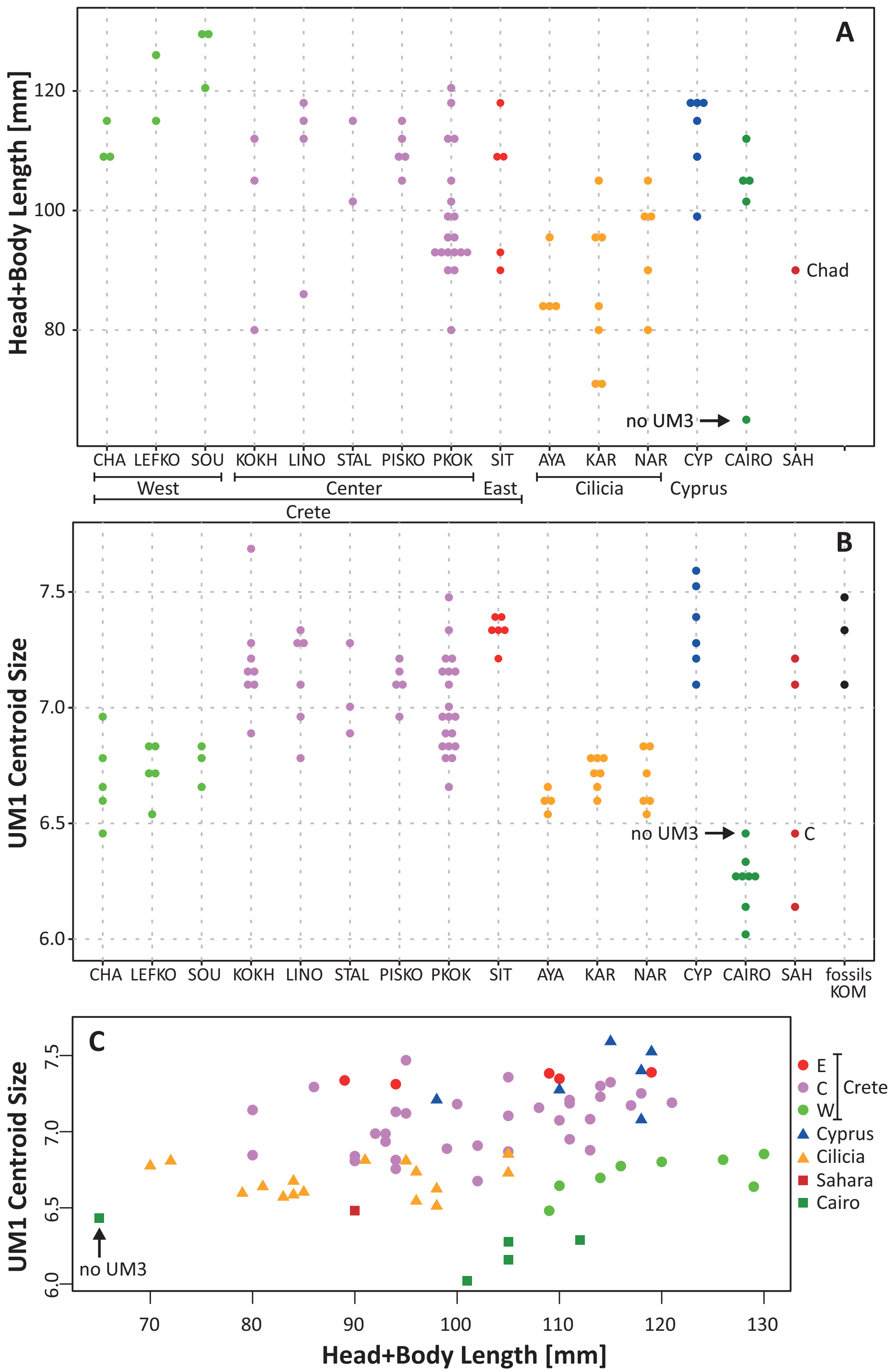

Figure 4. Size variation of *Acomys cahirinus s.l.* A, body size (head + body length) in modern populations from Crete, Cyprus, Turkey, Cairo and Sahara. B, size of the first upper molar (UM1) in the same populations; fossils from Kommos (Iron Age, Crete) are also included. C, relationship between body and molar size. The Chadian specimen among the Sahara sample is indicated by 'Chad' or 'C'. The arrow with 'no UM3' points to a young specimen without erupted third molar.

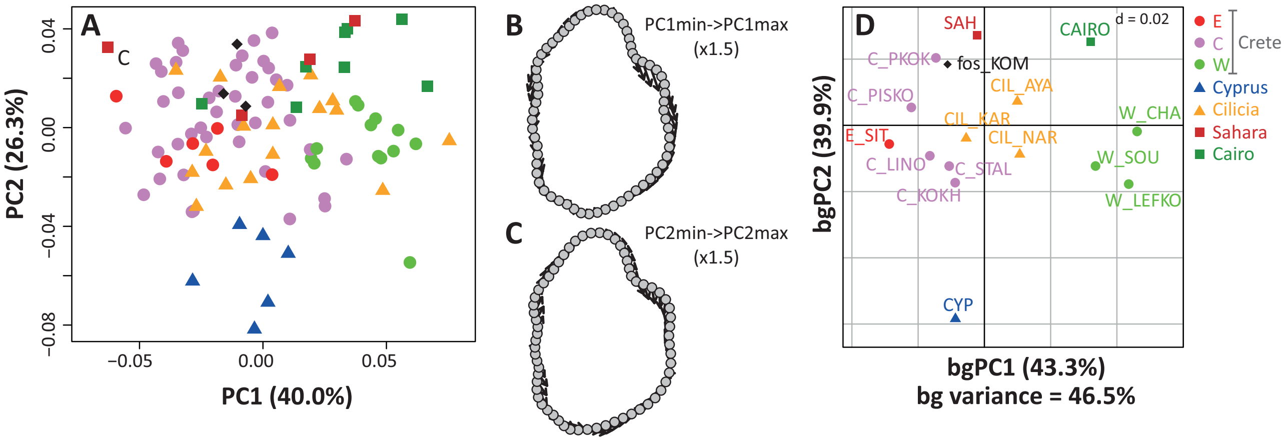

Figure 5. Shape differentiation of the first upper molar between the different populations of *Acomys cahirinus s.l.*, including the fossil teeth from Kommos (Iron Age, Crete). A, morphospace defined by the first two axes of a PCA on the aligned coordinates of the points delineating the outline of the first upper molar. Each dot corresponds to a tooth. 'C' indicates the Chadian specimen in the Sahara sample. B and C, deformation along the PC axes. Arrows point from the shape corresponding to the minimal score to the shape corresponding to the maximal score (B: PC1; C: PC2). D, means of the geographical groups in the morphospace defined by the first two axes of a bgPCA. Distance between grid bars: d = 0.02. For abbreviations see Table 1.

characterize teeth from Cyprus, with a broader and rounder posterior part, closer labial cusps (t3 and t9), an anteriorly shifted lingual cusp t1, and an anterior part compressed on the lingual side (Fig. 5C).

A bgPCA (Fig. 5D) provided further insight into the geographical clustering on Crete. The populations of Western Crete appeared to be the most differentiated. The Central and Eastern Cretan populations were relatively close to each other. The Cilician populations appeared central in this morphospace. Along the second axis of the bgPCA, Cyprus appeared highly divergent. At the opposite side of this axis, the two African samples shared high positive scores; the sample from Cairo was closer to Western Crete along bgPC1, whereas the Saharan sample was closer to Central and Eastern Crete. The populations from Central Crete varied along bgPC2, the two samples from Piskopiano and surroundings displaying positive bgPC2 scores, whereas the three other localities shared negative bgPC2 scores. The fossil teeth from Kommos plotted close to the samples from Sahara and Piskopiano in Central Crete.

Allometric tooth shape variation

Because tooth size variation appeared to be important (Fig. 4), allometry was investigated as a source of shape differentiation. Shape appeared to be significantly related to size (Procrustes ANOVA: P = 1e-04). The CAC based on this analysis (Fig. 6A) showed a general size–shape trend which may contribute to the shape similarity between the populations with small molars (Cairo, Western Crete) and to those sharing large molar size (Eastern Crete and Cyprus). Large teeth tended to display a prominent and anteriorly shifted first lingual cusp (t1) and a prominent posterior labial cusp (t9) compared to small teeth (Fig. 6B).

The size-free shape variation was explored based on a bgPCA on the residuals of a regression of the aligned coordinates on centroid size (Fig. 6C). The pattern of between-population differentiation was not deeply modified compared to the analysis on the raw aligned coordinates (Fig. 5D).

DISCUSSION

Two ancient haplogroups in Crete

This study confirms the complex geographical structure observed in previous studies within the Eastern Mediterranean spiny mice *Acomys cahirinus s.l.* (Barome *et al.*, 2001; Frynta *et al.*, 2010; Giagia-Athanasopoulou *et al.*, 2011). Different haplogroups occur in Cyprus and Cilicia, the only area where *A. cahirinus s.l.* occurs on a northern Mediterranean coast, while Crete hosts both haplogroups (Barome *et al.*, 2001; Frynta *et al.*, 2010). The divergence of the two haplogroups has been dated here at 210 kya, an estimate based on four genes, several calibration points within *Acomys* and a Bayesian approach, allowing us to re-evaluate the previous estimate of 400 kya (Barome *et al.*, 2001). Such a range of dating of divergence is typical for phylogeographical lineages isolated during the Plio-Pleistocene climatic fluctuations (e.g. Nicolas *et al.*, 2008; Ben Faleh *et al.*, 2012; McDonough *et al.*,

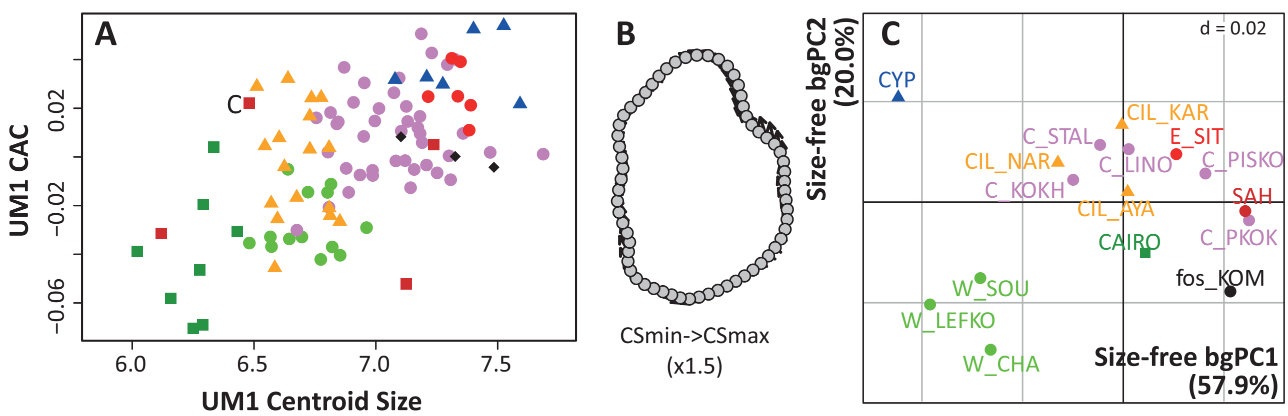

Figure 6. Allometric shape differentiation of the first upper molar. A, allometric shape variation. Size is estimated by the centroid size of UM1, and shape by the common allometric component (CAC). 'C' indicates the Chadian specimen in the Sahara sample. B, allometric deformation. Arrows point from the shape corresponding to the minimal centroid size to the shape corresponding to the maximal centroid size. C, means of the geographical group in the morphospace defined by the first two axes of a bgPCA on the residuals of a regression of the aligned coordinates vs. size. Distance between grid bars: $d = 0.02$.

2015). Because a Pleistocene fossil record of *Acomys* is absent from Cyprus and Crete (Barome *et al.*, 2001), the divergence between the haplogroups clearly predates the dispersion in this area.

A complex geographical structure in Cretan spiny mice

The two mitochondrial lineages have different geographical distributions: lineage A dominates in Western and Central Crete, whereas lineage B occurs mostly in Eastern Crete (Barome *et al.*, 2001; Frynta *et al.*, 2010; Giagia-Athanasopoulou *et al.*, 2011). The present study further shows a finer geographical structure, based on the repartition of the D-loop haplotypes: Western and Central Crete tend to display different haplotypes within lineage A (Fig. 3). This is in complete agreement with the morphometric results, showing different molar size (Fig. 4) and shape (Fig. 5) in Western, Central and Eastern Crete; possibly, a finer differentiation occurs even within Central Crete, with the area around Piskopiano displaying slightly different molar shape (Fig. 5). A reconsideration of the former chromosomal results (Giagia-Athanasopoulou *et al.*, 2011) also suggests regional variations in Crete (Fig. 7), with high, ancestral, diploid chromosome numbers (up to $2n = 42$) dominating in Central Crete, whereas lower, derived diploid chromosome numbers (e.g. $2n = 38$) are found in Western and Eastern Crete (Fig. 7). All markers, despite being very different in their nature (mitochondrial DNA, karyotypes, molar size and shape) are therefore congruent and point to a strong geographical structure in Crete.

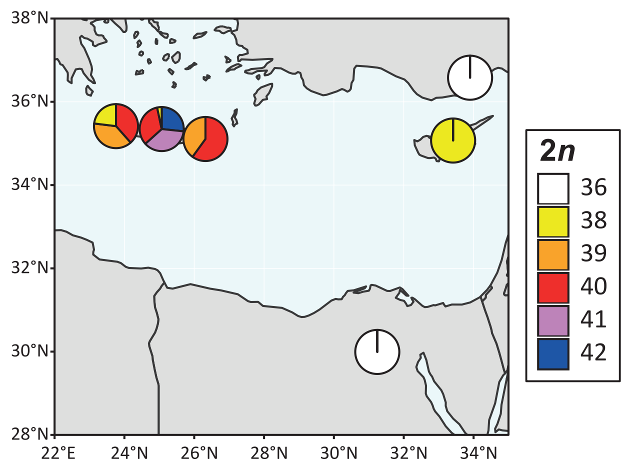

Figure 7. Karyotypic variation of *Acomys cahirinus s.l.* in the Eastern Mediterranean area. The pie charts indicate the proportion of the $2n$ numbers in the different populations. Data from Giagia-Athanasopoulou *et al.* (2011).

Nevertheless, evidence of mixing occurs. Regarding the mitochondrial lineages, specimens attributed to lineage B can occasionally be found in regions dominated by lineage A: one mouse from Akrotiri Peninsula, Chania (Barome *et al.*, 2001), and one mouse from the easternmost locality of Central Crete, Stalida Mochos (Giagia-Athanasopoulou *et al.*, 2011; this study). Regarding karyotypes, a specimen with the derived diploid chromosome number $2n = 38$ was found in Central Crete, despite the dominance there of high diploid chromosome numbers ($2n = 42$, $2n = 40$ and their hybrids). The occurrence in Eastern Crete of both homozygous mice with $2n = 40$ and heterozygotes with $2n = 39$ suggests the existence in the area of homozygous mice with $2n$ = 38, although they have not yet been captured (Fig. 7; Giagia-Athanasopoulou *et al.*, 2011).

Despite this evidence of mixing, the strong geographical structure observed for molar shape shows that gene flow is not sufficient to homogenize the populations across Crete. Crete is characterized by a mountainous relief compartmented by basins, with several massifs culminating above 2000 m. Both high mountains and basins may constitute barriers to dispersal for spiny mice preferably inhabiting Mediterranean environments with rocky substrates (Kryštufek & Vohralík, 2009). This complex geomorphology probably promoted isolation and divergence of *Acomys* populations in different parts of the island.

Cretan spiny mice within the diversity of the Eastern Mediterranean area

The diversity observed in Cretan spiny mice is difficult to place into a general pattern, because the diversity on the African continent remains largely under-sampled. However, the distribution of the two haplogroups in the Eastern Mediterranean area points to a complex history of migration. The haplotypes found in Crete, Cyprus and Cilicia were shown to be derived, compared to the haplotypes sampled so far in Libya, Chad and the high Nile Valley. Chadian and Libyan specimens even appeared to be outside of lineage A or B, due either to their basal position or to incomplete lineage sorting. In contrast, lineage A haplotypes sampled in Cairo appeared to be derived and branched with sequences from Cyprus and Western Crete (Frynta *et al.*, 2010; this study).

Regarding karyotype evolution, *Acomys* is characterized by an important variation of the diploid chromosome number, ranging from $2n$ = 68 to $2n$ = 40–42 and even $2n$ = 38–36 in the *A. cahirinus* group, given that $2n$ = 36 is the karyotype with the lowest chromosome number that can be achieved in spiny mice through Rb fusions (Lavrenchenko *et al.*, 2011). Within *A. cahirinus*, $2n$ = 40–42 karyotypes can therefore be considered as 'ancestral' and those of $2n$ = 36–38 as 'derived'. In particular, Central Crete hosts ancestral karyotypes ($2n$ = 40 and even $2n$ = 42). Cyprus displays an additional Rb fusion ($2n$ = 38), which is also present in Western Crete and possibly in Eastern Crete. A further Rb fusion has led to $2n$ = 36 in the Cilicia population and in the Egyptian population from Cairo (Macholán *et al.*, 1995; Zima *et al.*, 1999; Giagia-Athanasopoulou *et al.*, 2011). However, it is difficult to assess the variation within *A. cahirinus* in the African continent, due to limited geographical sampling and taxonomic uncertainty.

Regarding molar morphometrics, the population from Central Crete appears close to the Saharan group and to the fossil teeth from Kommos. Together, this suggests that Central Crete may host a population characterized by ancestral tooth morphology, karyotypes and haplotypes. The Cretan fossils were deposited during the lifetime of the Hellenistic temple of Kommos, which lasted from 375 BC to AD 160/170. Because no *Acomys* fossils have been found in older Cretan deposits (K. Papayiannis, pers. obs. January 2020), this places the earliest appearance of *Acomys* on Crete during this time interval.

The occurrence of a second mitochondrial lineage in Eastern Crete (lineage B), however, suggests that multiple introductions occurred from different continental source populations. Important trade across the Eastern Mediterranean area continued throughout the Bronze Age and Historical times (Karetsou *et al.*, 2001). The dispersion of *Acomys* through the Eastern Mediterranean area probably occurred by human-mediated transport (Barome *et al.*, 2001) as unintentional stowaways on boat cargos, in a manner similar to that described for the western house mouse *Mus musculus domesticus* (Cucchi, 2008). Both Cilicia and Cyprus spiny mice display more derived haplotypes, karyotypes and tooth shape than those from Crete; this further suggests that Crete acted as a hub from which spiny mice were translocated.

In support of this statement, the fact that the $2n$ = 38 karyotype of some Cretan spiny mice is identical to that of Cypriot mice, and differs only by an additional Rb fusion from the $2n$ = 36 karyotype of Cilician mice, could be explained through the expansion of mice with $2n$ = 38 from Crete to the other two regions, irrespective of mitochondrial lineages. The two lineages are reported to hybridize freely in the laboratory (Frynta *et al.*, 2010), and in Crete they are not associated with specific karyotypes (e.g. there exist spiny mice with $2n$ = 38 attributed to lineage A or B; Giagia-Athanasopoulou *et al.*, 2011). Thus, populations with $2n$ = 38 could have been imported to Cilicia from Eastern Crete (dominated by lineage B) and to Cyprus from Central or Western Crete (dominated by lineage A). The additional Rb fusion, mentioned above, would have then occurred locally in the isolated population of Cilicia, leading to its very derived karyotype ($2n$ = 36).

The population from Cairo constitutes a puzzling case. It displays the same derived karyotype as Cilicia ($2n$ = 36 with the same Rb fusions), but belongs to lineage A, with haplotypes related to those of Cyprus and Western Crete. Thus, a direct relationship between the Cairo and Cilician spiny mice is unlikely. Moreover, the tooth morphology of the Cairo mice closely matches in size and shape to that observed in Western Crete. These facts could be reconciled if the population from

Cairo is actually derived from a secondary import on the African continent, from Western Crete or Cyprus.

In this case, the identical karyotype of $2n$ = 36 in both Cilicia and Cairo would be the result of the independent, *in situ* fixation of the same Rb fusion in the karyotype of both populations. Derived from $2n$ = 38, there are only three acrocentric chromosomes available for a new Rb fusion, one of which is very small (Giagia-Athanasopoulou *et al.*, 2011). Rb fusions do not occur completely randomly between acrocentric chromosomes, but similarly sized chromosomes seem to be preferably fused (Gazave *et al.*, 2003), and this may have triggered the independent fixation of the same Rb fusion in the two, otherwise distant, populations (Lavrenchenko *et al.*, 2011). As in the house mouse, successive Rb fusions, leading to very low diploid numbers, might have been favoured by successive dispersion events, and a small patchy distribution (Auffray, 1993). The peculiarity of the Cairo population could be maintained by behavioural mechanisms, similar to those reported for house mouse populations from Tunisia (Chatti *et al.*, 1999), as this population is characterized by its commensal habit (Kryštufek & Vohralík, 2009) that may maintain its isolation from the surrounding populations. Any further interpretation is hampered by the limited data available from the African continent.

Insularity and isolation promoting morphological divergence

Whatever the dynamics of colonization, the insular conditions promoted a pronounced morphological divergence in tooth morphology. This occurred in a relatively short evolutionary period, since the first appearance of *Acomys* in Crete occurred during the Hellenistic period, ~2000 years ago (Payne, 1995). The evolution on Cyprus is even more rapid. As no *Acomys* fossil has been recovered so far from the prehistoric record (Vigne, 1999; Horwitz *et al.*, 2004), the species is thought to have been introduced to this island during the last 1000 years. In that period, the molar morphology evolved markedly, exemplifying the acceleration of morphological evolution on islands (Millien, 2006). Morphological divergence occurred as well in the isolated but not insular population of Cilicia, although to a lesser degree. This suggests that random drift in isolated, small populations fostered rapid divergence in Crete, Cyprus and Cilicia, and that adaptation to local insular conditions has further driven molar evolution in the two insular populations, as shown for the house mouse (Ledevin *et al.*, 2016). In addition, hybridization between the populations on Crete might have contributed to the diversity of molar shape on this island. Hybrid morphologies are not necessarily intermediate between those of the parents, but can display transgressive phenotypes (Renaud *et al.*, 2017a). In Central Crete, molars frequently display an elongated forepart up to the occurrence of a small, additional cusplet in front of the t2 cusp. A similar phenotype has been found in hybrid house mice (Renaud *et al.*, 2017a), and in several insular house mouse populations (Renaud *et al.*, 2011, 2018). The evolution of similar dental phenotypes in distant rodent groups might be due to shared developmental processes favouring not only convergent evolution within true murines (Hayden *et al.*, 2020), but also between murines and the murine-like molar of *Acomys*.

Heterogeneity in molar size: a partial decoupling from body size

Evolution of size is a well-known characteristic of insular populations (Lomolino, 1985, 2005). Small mammals are expected to increase in body size, due to a combination of factors including decreases in interspecific competition and predation pressure, and increase in intraspecific competition (Lomolino, 1985, 2005). Because molar size is considered to be a good proxy of body size at a broad taxonomic scale (Gingerich *et al.*, 1982), an increase in molar size could be expected as well.

Regarding body size, *Acomys* indeed displays a large body size on Cyprus, but within Crete, body size varies considerably among populations (Fig. 4A). This may be related to the different age structure of the populations: sampling at different periods may lead to an overrepresentation of young or old animals, skewing the size distribution towards small or large body size (Renaud *et al.*, 2017b). Such a confounding factor may be especially important in a species in which neonates already display hair and open eyes, and can therefore rapidly leave the nest. Age structure may also contribute to the large body size observed for most specimens from Cairo, which are noted to have been maintained in captivity for some time.

Estimating the degree of insular size increase is made more difficult due to the lack of sufficient data for African spiny mice. Nevertheless, among Eastern Mediterranean spiny mice, the continental Cilician *Acomys* displayed the smallest body size. Differences between Cypriot and Cretan mice may be related to local differences in competition and predation. For instance, the least weasel (*Mustela nivalis*) has been intentionally introduced by Phoenicians and Greeks on the Mediterranean islands, with the aim of controlling commensal rodents following their stowaway introduction (Rodrigues *et al.*, 2017). The weasels introduced in antiquity (Lehmann & Nobis, 1979), however, went extinct in Cyprus (Rodrigues *et al.*, 2017), further relieving mammalian predation pressure on Cypriot *Acomys*.

In contrast, molar size was found to vary greatly among Crete, Cyprus and Cilicia, and even within Crete. This apparently surprising uncoupling between molar and body size is due to the fact that even if molar size is correlated to body size at a broad taxonomic scale, it is not necessarily the case at a population level, because the first molar erupts early after birth and is therefore not affected by subsequent growth (Renaud *et al.*, 2017b). The most obvious decoupling between body and molar size is found for populations from Western Crete and Cairo, being of relatively large body size but extremely small molar size (Fig. 4C). Similar uncoupling is not uncommon and has been observed for instance in wood mice (*Apodemus sylvaticus*) from Ibiza (Renaud & Michaux, 2007).

The large variation in molar size may echo an under-evaluated variation in the African continent, as suggested by the large variation in tooth size observed in Khartoum, Sudan (Fig. 4B). It could also have an adaptive component, for instance related to a polarity in rainfall in Crete, western regions receiving more rainfall and thus hosting different ecosystems. Indeed, larger tooth size ('macrodonty') has been proposed to occur in some insular small mammals as a response to diet shifts (Vigne *et al.*, 1993). Large teeth with a massive outline, as with those of Eastern Crete, might be favoured in dry habitats, in order to process more efficiently hard food that is resistant to comminution (Renaud *et al.*, 2005).

CONCLUSION

Our genetic, chromosomal and morphometric results were congruent with a strong geographical structure in Eastern Mediterranean spiny mice (*Acomys cahirinus s.l.*) and especially within Crete. A complex history of multiple introductions is probably responsible for this structure. Insular isolation coupled with habitat shift must have further promoted a pronounced and rapid morphological evolution in molar size and shape on Crete and Cyprus.

As a consequence, the species *A. nesiotes* (Cyprus), *A. cilicicus* (Cilicia, Turkey) and *A. minous* (Crete) are clearly nested within *A. cahirinus* (group cah9 in Aghová *et al.*, 2019) and are only defined by their geographical distribution. Due to their isolated distribution and morphological characteristics, the names '*nesiotes*' and '*cilicicus*' may be maintained for describing subspecific evolutionary units. *Acomys minous* encompasses populations with different haplotypic composition and morphometric characteristics, and therefore it does not correspond to a homogeneous evolutionary unit. More thorough sampling of the African continent, as well as further genetic data including nuclear genes, would be required for a better understanding of the complex history of translocation and evolution in isolation that led to the amazing morphological diversity in modern Cretan and, more generally, Eastern Mediterranean spiny mice.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Accessions of the sequences used in the dating analysis. Lineages and groups are defined as in Aghova *et al*. (2019).

Table S2. Substitution models used in the BEAST analysis.

Figure S1. Chronogram obtained with BEAST. Values at nodes represent medians of estimated divergence date, and the horizontal bars show the 95% highest posterior density of these estimates. Red stars indicate the positions of three fossil constraints used for calibration of the molecular clock.